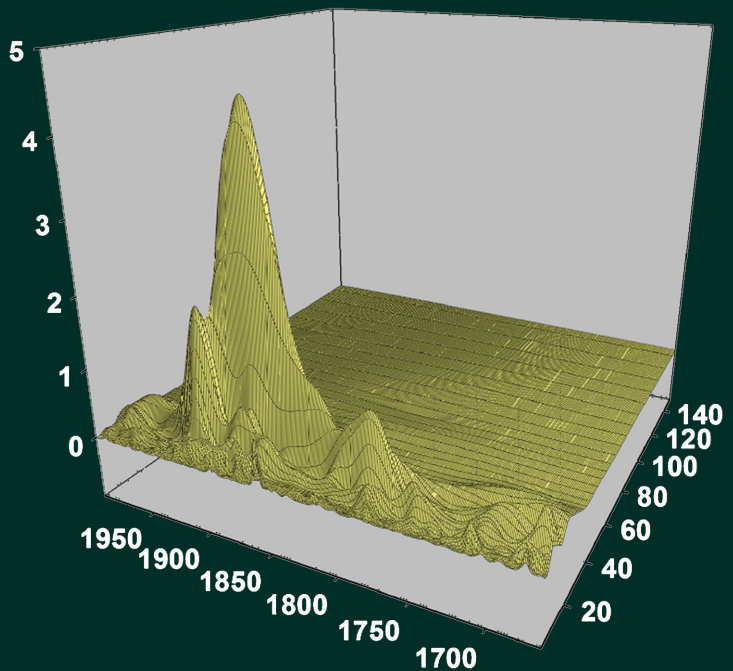
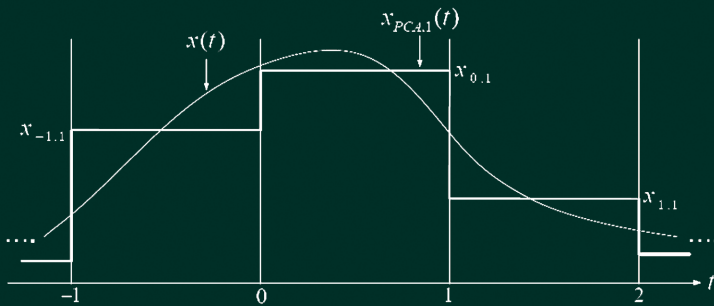


Wavelets and Fractals in Earth System Sciences



Editors

E. Chandrasekhar

V. P. Dimri

V. M. Gadre



CRC Press
Taylor & Francis Group

A CHAPMAN & HALL BOOK

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in
Earth System Sciences

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Boca Raton London New York

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Version Date: 20131017

International Standard Book Number-13: 978-1-4665-5360-6 (eBook - PDF)

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Foreword

Wavelets were invented by the French geophysicist Jean Morlet, so it is most appropriate that the Department of Earth Sciences at the Indian Institute of Technology Bombay (IITB), with support from the Ministry of Earth Sciences and the Department of Space, organized an interesting meeting on wavelets and fractals in earth sciences at IITB early last year. The story of Morlet's invention is well known, and has been narrated in tributes given by Yves Meyer and Pierre Goupillaud. The wavelet idea had some precursors in the form of techniques to provide information on frequency scales in running time, as the human ear does so naturally in music and speech for its owner. An example of the kind of problem that such techniques attempt to handle is the decomposition of a signal whose frequency is constant over relatively short times but otherwise keeps varying throughout (as it happens familiarly in Indian music). Classical Fourier series cannot handle such inherently nonstationary phenomena. Morlet's achievement consisted of devising a system that was most appropriate for this class of problems; a 2-D transform of a 1-D transient, nonstationary signal could be defined, and furthermore, the relation could be inverted to reconstruct the original signal from the transform. The signals that Morlet was handling were those encountered in prospecting for oil, which was the business run by his employer; here, the waves transmitted into and reflected back from different layers in the soil return to the surface, with different frequencies at different times. Morlet's first article on the subject seemed so far-out and abstract to the geophysical journal he submitted it to that it was rejected. However, a collaboration with a theoretical physicist, Alex Grossmann, led to an elegant formulation that was mathematically convincing and sharply crystallized the wavelet idea.

Since then, work on wavelets has grown very rapidly. As often happens with new ideas, the first reactions of skepticism to Morlet's work were quickly followed by great appreciation after the publication of the Morlet-Grossmann article in 1984. There is now a whole series of wavelets of different kinds for different applications that range from data compression to image processing, and the detection of hidden order in apparently chaotic signals. Many books are now available on the subject—from relatively abstract mathematical treatments to books such as *Wavelets for Dummies*. The technique has been used not only in seismology and oil prospecting but also in meteorology, fluid dynamics, quantum physics, data analysis and compression, characterization of archeological structures or volcanic activity, and so on, to multiscale filtering and processing in electrical engineering. There are discrete and continuous transforms, 1-D, 2-D, and 3-D transforms, those that are good at detecting sharp lines and others in smoothing them out. The list is long.

This volume covers a great deal of this ground. Although it has a legitimate emphasis on geophysical applications, there are also chapters by electrical, aerospace, and materials engineers. The revolution that wavelets have wrought is that they provide, for the first time, a set of versatile tools that would be the default choice if today one is investigating transient non-stationary processes or data of any kind. This book should help to bring out the richness of wavelet (and fractal) methods now available, and introduce them to new entrants to research by offering a good mix of theory and specific applications in many different fields.

I wish to compliment the Indian geophysical and engineering communities for getting together to organize this very useful, timely and stimulating meeting.

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Preface

The subject of wavelet analysis and fractal analysis is fast developing and has drawn a great deal of attention in varied disciplines of science and engineering. Over the past couple of decades, wavelets, multiresolution analysis, and multifractal analysis have been formalized into a thorough mathematical framework and have found a variety of applications with a significant effect on several branches of earth system sciences, such as seismology, well-logging, potential field studies, geomagnetism and space magnetism, atmospheric turbulence, space-time rainfall, ocean wind waves, fluid dynamics, seafloor bathymetry, oil and gas exploration, and climate change studies among others. It is certain that there will be plenty of applications of wavelets and fractals in earth sciences in the years to come.

This book primarily addresses the important question “Why *not* another book?” instead of “Why another book” on wavelets and fractals? Commensurate with the rapid progress in the application of wavelets and fractals in all fields of science and engineering in general, and in earth sciences, in particular, the available books and other forms of literature on these novel signal analysis tools are alarmingly few. This has been the motivation for us to edit this book. Through this book, we attempt to highlight the role of such advanced data processing techniques in present-day research in various fields of earth system sciences. The book is composed of a unique collection of a wide range of application-oriented research topics in a multitude of fields in earth system sciences and presents the same under one umbrella.

The book consists of 10 chapters, providing a well-balanced blend of information about the role of wavelets, fractals, and multifractal analyses with the latest examples of their application in various research topics. We admit that the reader may find some basic concepts of continuous and discrete wavelet transformation techniques being repeated in some chapters. Given the different perspectives and wide applications of these techniques in different fields of science, it was felt necessary to retain this repetition for the sake of continuity in the text, particularly when such information is important and relevant to the research content of the chapter, for its better understanding. The chapters are written by experts in their respective fields. Starting from the fundamental concepts of the theory of wavelets and fractals, their application in a variety of fields in earth sciences is described. The construction of wavelets, second-generation wavelets, the role of fractals in earthquake prediction, the application of wavelets in geomagnetism, potential field studies, atmospheric rainfall anomalies, paleoclimate studies, and phase field modeling constitute the book.

In Chapter 1, the fundamental concepts of wavelets and fractals and their use in various branches of geosciences research are discussed albeit without

many of the mathematical details. The concept of multifractals and their necessity for the effective understanding of complex geophysical phenomena such as earthquakes is also explained in this chapter. Finally, the relation between wavelets and fractals and the advantage of wavelet-based fractal analysis in studying the signals is discussed.

Chapter 2 builds up the whole philosophy of filter banks as a template for the construction of wavelets. Beginning with the Haar wavelet and using that as a platform to explain the close relation between filter banks and wavelets, the authors explain the need for “higher order” filter banks in wavelet analysis. Besides explaining the construction of some classes of orthogonal and biorthogonal wavelets, the authors present some of their own approaches for designing filter banks keeping the uncertainty or time–frequency bandwidth product as a criterion for optimization. Some designs are presented along with their performance in a compression application, which is very relevant to solving problems in the field of earth sciences.

Chapter 3 presents the genesis of wavelet transform. It discusses the development of second-generation wavelets and their construction. A comprehensive review of some applications of wavelet transform in the field of geophysics, particularly in potential fields, is highlighted.

Chapter 4 discusses multiresolution decomposition as a way to create multiscale relationships that help to identify the self-similarity and singularity properties in the data. It emphasizes the point: Why is multiscale representation very useful in studies such as data compression and noise removal? The role of wavelets in data compression is especially useful to store large volumes of geological maps, subsurface geophysical models, and remote sensing images.

Chapter 5 discusses some examples of the application of fractals and wavelets in geophysics. The most generally used “box-counting” method to determine the fractal dimension is detailed with an illustrative example of field data. An important application of wavelets to reduce the runtime for solving large matrices in gradient-based inversion of geophysical data is explained, as well as providing an example of the application of wavelets in analyzing potential field data.

Chapter 6 highlights the application of multifractals in earthquake prediction studies. Temporal variations of heterogeneity in seismicity using multifractal analysis have been reported in various tectonic regions for both microseismicity as well as macroseismicity data. By examining the generalized fractal dimension Dq and the spatiotemporal variation of Dq spectra, representative of seismicity pattern and their distribution in any region, the role of multifractals in earthquake prediction is highlighted.

Chapter 7 explains the role of wavelets in the field of geomagnetism. It introduces complex wavelets and their application in understanding the phase characteristics of geomagnetic jerks, which occur due to the differential fluid flow on the surface of the outer core. This study, carried out using global magnetic observatory data from approximately eight decades, helps

in understanding their space–time and time–frequency localization characteristics. Geomagnetic jerks, having a period of about one year, manifest sudden changes in the slope of geomagnetic secular variations. Phase characteristics of geomagnetic jerks aid in better understanding the direction of outer core fluid motions responsible for the generation of geomagnetic jerks. This chapter also highlights the hemispherical differences observed in the nature of occurrences of geomagnetic jerks.

A unique application of wavelets in studying the paleomonsoon data from speleothems is detailed in Chapter 8. Stable oxygen isotope variations in cave calcite (CaCO_3 , stalactites and stalagmites), collectively known as speleothems, are used for paleomonsoon reconstruction. In this chapter, the authors present wavelet transformation together with multitaper analysis of the same data, and explain how the use of different filters can help identify hidden signals pertaining to solar modulation of the monsoon.

Another unique application of wavelets is in studying tree-ring data that is helpful for understanding rainfall anomalies and climate studies, which is detailed in Chapter 9. In this chapter, the authors highlight the different periodicities that they have obtained in their analysis of rainfall data, and tree-ring data and the good correlation among them. They also discuss insights on their effect on various present-day global phenomena.

Application of wavelet transformation in the studies of the kind presented in Chapters 8 and 9 are very rare and thus this book enjoys the privilege of having such rare discussions on such important areas of research, which are beneficial for monsoon and climate change studies.

Finally in Chapter 10, another unique application of wavelet transformation, namely, that of solving nonlinear partial differential equations (specifically, those from a phase field model), is presented. Phase field models are widely used in the field of materials science and engineering to study microstructures and to understand the solid–solid and solid–liquid boundaries. By carrying out a suitable nondimensionalization, these models can also be used to study boundaries at macro scales such as crust–mantle (solid–solid) and mantle–core (solid–liquid) boundaries. In this chapter, the authors describe how phase field models based on the Cahn–Hilliard (CH) equation can be used to understand the phase transition dynamics and the resultant (stress assisted) microstructural evolution of the interface between a non-hydrostatically stressed mineral and its solution. They also present results from a Fourier spectral implementation for the two-dimensional CH equation (with elastic stress effects), and a wavelet-collocation implementation for the one-dimensional CH equation (without elastic stress effects).

Some key features of the book are

- Prior knowledge of wavelets and fractals is not required to understand this book because each chapter introduces the needed concepts of wavelets and fractals.

- Focused set of applications centered on themes of thrust areas of present-day research in science and engineering.
- Healthy combination of basics and advanced material.
- Serves as excellent introductory material and also as an advanced reference text for students and researchers.

We place on record our sincere thanks to the Ministry of Earth Sciences, Government of India, ISRO (GBP) and ISRO (RESPOND) of the Department of Space, Government of India, for financial support. We express our sincere thanks and deep sense of gratitude to Prof. Roddam Narasimha, FRS, for readily agreeing to write the Foreword for this book and for his timely advice and moral support. We also express our sincere thanks to all the contributors of the chapters and the reviewers for helping us with their constructive criticism and meticulous reviews of the chapters. Finally, we thank the staff of CRC Press, Taylor & Francis group, particularly, Aastha Sharma, Marsha Pronin, and Rachel Holt and others for their kind cooperation and support throughout the evolution of this book.

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1

Introduction to Wavelets and Fractals

E. Chandrasekhar and V. P. Dimri

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Introduction

Wavelets

The early 1800s saw a revolution in the understanding of mathematical functions when Jean Baptiste Joseph Fourier, in his groundbreaking observations, proclaimed that any mathematical function can be represented as a combination of several sines and cosines. This way, because the underlying frequencies of sine and cosines were known, it became easier to distinguish the frequencies of interest present in the signal under investigation. With the advent of subsequent thorough mathematical formalism for this transformation, the Fourier transform (FT) came into existence and is still used extensively to solve a variety of problems in science and engineering.

The sine and cosine functions are the basis functions in Fourier theory, which for any frequency is infinitely oscillatory in time and thus is *not* compactly supported (i.e., mathematically, the Fourier basis functions do not belong to the $L^2(\mathbb{R}^n)$ functional space). As a result, Fourier theory can determine only the presence of all frequencies in the signal and cannot estimate when in time these frequencies of interest occur (also known as time localization). For example, if you have three signals in time-domain, with the first signal having a low-frequency part followed by a high-frequency part (Figure 1.1a), the second signal having a high-frequency part followed by a low-frequency part (Figure 1.1b), and the third signal is a combination of both these frequencies (Figure 1.1c), then the Fourier transformation in all three cases will yield the same spectra (barring the small ripples that you observe in the first two cases in their FT spectra that arise because of Gibbs phenomenon due to sudden frequency transitions; Figure 1.1). Now, the question is, how can the spectra of three different signals be the same? The answer is, this is the limitation of FT. This explains that FT can only tell us what frequencies are present in the signal and their respective average amplitude. Nothing more, nothing less. In other words, with FT, only the frequency localization is possible, but not the time localization.

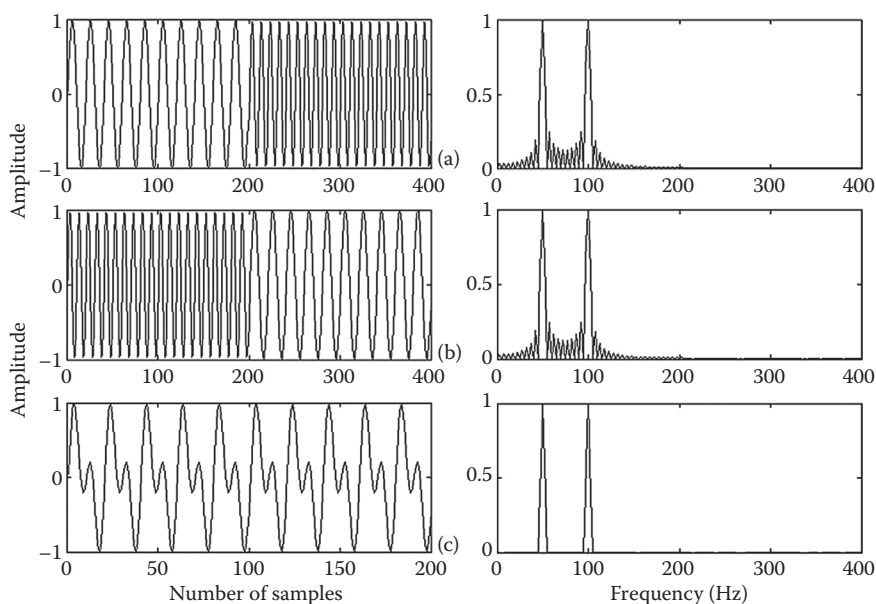


FIGURE 1.1

Example of three different signals with similar Fourier spectra. The left panels show (a) low-frequency signal appended with its high-frequency counterpart, (b) high-frequency signal appended with its low-frequency counterpart, and (c) a combination of both frequencies shown in (a) and (b). Their respective Fourier spectra are shown in the right panels.

However, it is becoming increasingly important and necessary to study some signals based not only on the frequency content but also on their time of occurrence (also called time–frequency localization) for a correct understanding and interpretation of the physical processes that generate such signals.

For example, knowledge of the behavior of the Earth's magnetic field variations at different times over different regions is always important to understand the spatiotemporal behavior of certain frequency characteristics of these variations (see Chapter 7 of this book and references therein). Similarly, in geophysical well-log data analysis, because different subsurface rock formations have different physical properties, which in turn reflect respective frequency characteristics, it becomes important to identify the locations of these frequencies (read formations) as a function of depth (space–frequency localization), particularly if these formations are oil-bearing/gas-bearing zones (see for example, Chandrasekhar and Rao).

Recognizing the drawback of FT, Gabor (1946) developed its small variant, called windowed FT (WFT) or short-time FT. The underlying principle of this technique is to break up the signal into smaller sections with fixed window length and Fourier analyze each section for its frequency content. Mathematical representation of the WFT of a time function $f(t)$ and a window function $g(t)$ is given by

$$\text{WFT}_g(\tau, s) = F_g(\tau, s) = \int_{-\infty}^{\infty} f(t)g(t - \tau)e^{-i2\pi st} dt \quad (1.1)$$

$F_g(\tau, s)$, which is essentially the FT of $f(t)$ and $g(t - \tau)$, represents the amplitude and phase of the signal over time and frequency. $|F_g(\tau, s)|^2$ gives the spectrogram of $f(t)$. In WFT, although the short lengths of data are considered at each step, the main transformation again is FT, whose basis functions, as mentioned above, are infinitely supported and therefore the problem of time–frequency localization still persists. Furthermore, because the window length is fixed at each step in WFT, the resolution in the entire time–frequency plane will be the same (see Graps 1995; Mallat 1999). This is the drawback of WFT. Alternatively, although a wider window gives good frequency resolution, a narrower window gives good time resolution (Polikar 2001), indicating that they are governed by the uncertainty relation. Hence, WFT also was found not to be suitable where time–frequency or space–frequency localization is important.

Later, Banks (1975), with his complex demodulation technique, came out with some improvements in time–frequency localization of long-period geomagnetic data. This technique helped to determine the instantaneous power levels of the required frequency in the selected frequency band as a function of time. These are known as demodulates. Because each

demodulate represents the time-local estimate of amplitude and phase of selected frequency in the entire data sequence, they can be treated as independent entities and thus it becomes very useful to examine each demodulate separately for their time–frequency characteristics in a given time series (Chandrasekhar 2000). Despite providing the instantaneous power spectrum and phase, the main drawback of this technique is that it does not have constant Q frequency resolution and also lacks a thorough mathematical framework (Zhang and Paulson 1997). The Q factor of a window (also called the fidelity factor) defines the ratio of center frequency of the filter to its bandwidth.

The uncertainty relation explains that, to have a better resolution of time and frequency, there must be a window function defined whose width should be allowed to vary inversely with the frequency. This facilitates in having a constant Q frequency resolution, and thus the wavelet is born! Oscillatory short waves, having finite amplitude and finite time duration, are known as *wavelets*. These time-limited waves can be successfully utilized to analyze the signals under investigation and provide not only the effective time–frequency localization of signals but also identify the hidden discontinuities or spikes of interest in the given signals. Wavelets thus facilitate the correct interpretation and understanding of the spatiotemporal behavior of various signals, which otherwise, is not possible with the FT.

Fractals

Fractal theory was not invented in a day by Benoit B. Mandelbrot, who coined this term to open a new discipline of mathematics that deals with nondifferentiable curves and geometrical shapes. Until Karl Weierstrass presented his work at the Royal Prussian Academy of Sciences on July 18, 1872, mathematics was confined to the study of differentiable functions. Weierstrass showed that for a positive integer a and $0 < b < 1$, the analytical function

$$\sum_{n=1}^{\infty} b^n \cos(a^n \pi x) \quad (1.2)$$

was not differentiable.

There are a few others, whose works are closely related to fractal theory, such as Von Koch, Julia, Fatou, and Hausdorff, who laid the foundations that were generously exploited by Mandelbrot to establish a fascinating branch of mathematics to analyze the natural geometrical shapes and more precisely quantify the length, area, and volume of these shapes. It was a landmark essay entitled, “How long is the coast of Britain? Statistical self-similarity and fractional dimension,” written by Mandelbrot (1967), in which he related the work of his predecessors to real-world problems, namely, coastlines,

clouds, mountains, and many other natural shapes, which he claimed are self-similar* and devised a method to quantify the geometrical properties of these shapes with the help of a dimension, which he termed as the fractal dimension. It is important to mention that as stated above, the fractal theory provides a framework to study irregular shapes as they exist, instead of approximating them using regular geometry, as has been done hitherto. For example, earth is approximated as a sphere, coastlines are approximated with straight lines, and many other natural shapes to the corresponding nearby (or closer) regular geometry, simply for the sake of mathematical convenience.

The concept of fractals is not confined to the study of geometrical shapes, as it might seem from the above introduction, rather it has been extended to the analysis of numerical data, for example, the variation of some physical properties of the earth with depth, spatial, and time series data from various branches of science including complex and extreme events (Sharma et al. 2012).

In the following sections, we provide a brief description of the mathematical concepts of wavelets and fractals, and their superiority over the existing signal processing tools, particularly, FT. We briefly discuss the methodology of continuous and discrete wavelet transformation and fractal analysis, whereas more details of these techniques are given in other chapters. Finally, we discuss the geophysical significance of these novel data analysis tools.

Basic Theory and Mathematical Concepts of Wavelets and Fractals

Wavelets

The word *wavelet* refers to a small wave, having finite length in space (ideally, functions representing such waves will be zero outside a finite time interval; also known as *compactly supported*) and is represented as a real-valued function, $\psi(t)$, adhering to the following conditions (Percival and Walden 2000)

$$\int_{-\infty}^{\infty} \psi(t) dt = 0 \quad (1.3)$$

and

* If an object retains its original shape even after decomposing into smaller parts, then that object is said to have self-similar characteristics and is a fractal. Examples: Cantor set, Julia set, Cauliflower, cobweb, etc.

$$\int_{-\infty}^{\infty} \psi^2(t) dt = 1 \quad (1.4)$$

Equation 1.3 defines the *admissibility condition*, which explains that the wavelet function, $\psi(t)$, must be oscillatory in time and that it integrates to zero (i.e., having zero average). Equation 1.4 defines the *regularity condition*, which explains that the wavelet function, $\psi(t)$, must be finite in length and is square integrable (having finite energy). The wavelet function, $\psi(t)$, signifying the time–frequency localization is given by

$$\psi_{\tau,s}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right) \quad (1.5)$$

where s (>0) indicates the scale and τ indicates the translation parameter. Equation 1.5 facilitates the provision to allow the width of the window function s to vary inversely with the frequency. Accordingly, s is analogous to frequency, in the sense that higher scales correspond to the low frequency content of the signal and lower scales correspond to high frequency content of the signal. The frequency–scale relation is given by $f = \frac{f_c}{s\Delta t}$, where f_c denotes the central frequency of the wavelet, Δt the sampling interval, and f the frequency corresponding to the scale. The translation parameter τ is linked to the time location of the wavelet function. As the wavelet is dilated and shifted during its operation on the signal (see subsections on Some Properties of Wavelets), it provides time-scale information in the transformed domain. A number of wavelets satisfy the conditions prescribed in Equations 1.3 and 1.4, and thus they form what is called “wavelet families.” The function $\psi(t)$ is called “analyzing wavelet” or “mother wavelet.” More details about the fundamentals of wavelet theory can be found in Daubechies (1992) and Mallat (1999, and references therein).

The continuous wavelet transformation (CWT) of a function, $f(t)$, is the result of its inner product with the wavelet function (Equation 1.3), given by

$$\text{CWT}_{\tau,s} = \frac{1}{\sqrt{s}} \int f(t) \cdot \psi\left(\frac{t-\tau}{s}\right) dt \quad (1.6)$$

where, $f(t), \psi(t) \in L^2(R)$. Equation 1.6 explains that the wavelet transformation gives a measure of the similarity between the signal and the wavelet function. Such a measure at any particular scale s_0 and translation τ_0 , is identified by a wavelet coefficient. The larger the value of this coefficient, the higher the

similarity between the signal and the wavelet at (τ_0, s_0) and vice versa. Higher wavelet coefficients indicate the high degree of suitability of the wavelet to study the signal. Another important point to note here is that, when s is close to zero, the CWT coefficients at that scale characterize the location of the singularity (if present) in the signal, in the neighborhood of τ . Central to this is its application to detect sudden jumps (transients) and/or discontinuities in the data and analyze fractals (see Mallat 1999).

From Equation 1.6, it can be understood that the CWT of a one-dimensional function is two-dimensional. Furthermore, it can also be shown that the CWT of a two-dimensional function is four-dimensional, and so on.

Some Properties of Wavelets

Vanishing Moments

The regularity condition (Equation 1.4) imposes an additional constraint on the wavelet function and makes the wavelet transform decrease quickly with decreasing s . This can be better explained with the concept of vanishing moments of the wavelet. If the wavelet coefficients for m th order polynomial are zero, then that wavelet is said to have m vanishing moments. This means that any polynomial up to order $m - 1$ can be represented entirely in scaling function space. In other words, a wavelet has m vanishing moments if its scaling function (also known as “father wavelet”) can generate polynomials to a degree $\leq m - 1$ (see section on Discrete Wavelet Transformation for more details about scaling function). Alternatively, another simple way of defining wavelet vanishing moments is if the FT of the wavelet $\psi(t)$ is m times continuously differentiable, then the wavelet is said to have m vanishing moments (see Chandrasekhar and Rao 2012). Together with regularity condition, vanishing moments of the wavelet decide the fast decay of the wavelet.

Compact Support

This property explains that the wavelet vanishes outside a finite time interval. Shorter intervals indicate higher compactness of the wavelet and vice versa.

Translational Invariance

Another important property of wavelets is the translational invariance property, which explains that even a small time shift in the wavelet function results in a corresponding shift in the CWT output. Let us examine this.

Let $f_{\delta t}(t) = f(t - \delta t)$ be the translation of $f(t)$ by a small time shift, δt . The CWT of $f_{\delta t}(t)$ is

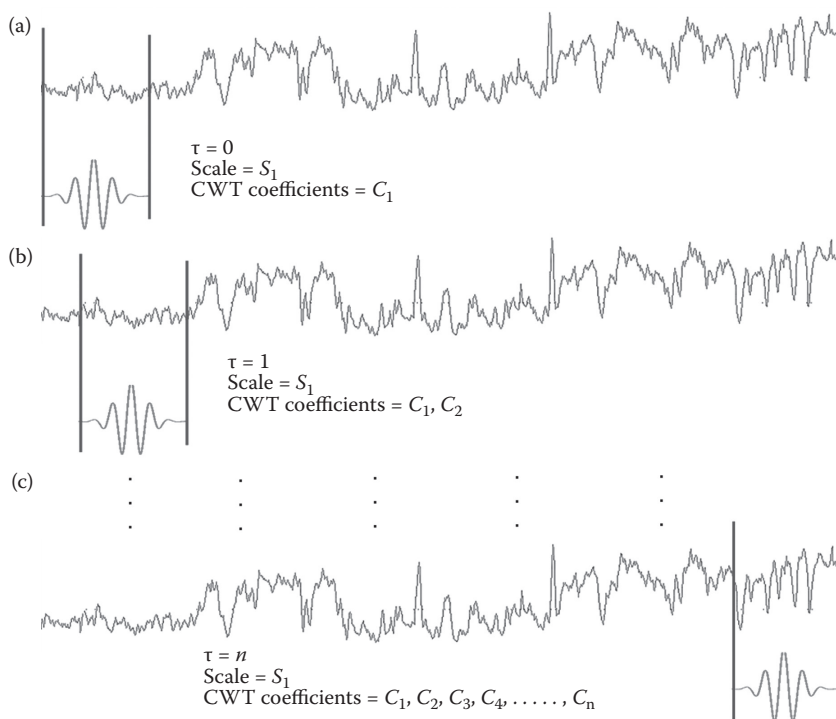
$$\begin{aligned}
\text{CWT } f_{\delta t}(\tau, s) &= \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} f(t - \delta t) \psi\left(\frac{t - \tau}{s}\right) dt \\
&= \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} f(t') \psi\left(\frac{t' - (\tau - \delta t)}{s}\right) dt \text{ where } t' = t - \delta t \quad (1.7) \\
&= \text{CWT } f(\tau - \delta t, s)
\end{aligned}$$

Since the output is shifted the same way as the input signal, the CWT is translation-invariant. Similarly, the shift-invariant property for discrete wavelet transform (DWT), in which the shifts are in dyadic scales can also be shown. For further details, the reader is referred to Ma and Tang (2001).

The discussion that we have had thus far on wavelets directly relates to the information that you will find in the following chapters. Hence, we limit our discussion to these aspects only. More details about different types of wavelets, their properties and applications, can be found in the respective chapters that follow. The reader is also advised to look into some fundamental books by Mallat (1999), Daubechies (1992), Chui (1992), Kaiser (1994), and Percival and Walden (2000), to cite a few.

Continuous Wavelet Transformation

In CWT (Equation 1.6), the signal to be transformed is convolved with the mother wavelet and the transformation is computed for different segments of the data by varying τ and s . Because the wavelet window can be scaled (shrunk or expanded) at different levels of analysis, the time localization of high-frequency components of the signal and frequency localization of low-frequency components of the signal can be effectively identified. In wavelet analysis, the wavelet window is first placed at the start of the signal. The signal and wavelet are convolved and the CWT coefficients are estimated. Next, the wavelet is translated (shifted in time) along the signal by a small amount and the CWT coefficients are again computed. This process is repeated until the end of the signal is reached. Next, the wavelet window is dilated (or stretched) by a small amount and placed at the start of the signal and the above process is repeated at that scale, at all translations. Likewise, wavelet spectra are calculated for many scales. Figure 1.2 describes the pictorial representation of the above operation. Finally, all the CWT coefficients computed at different dilations and translations are expressed in the form of scalograms, explaining the time-scale representations of the signal. Because of this, the wavelets are known as “mathematical microscopes.”

**FIGURE 1.2**

Pictorial representation of stepwise calculation of CWT coefficients. First, the CWT coefficients, when the wavelet with a translation $\tau = 0$ and dilation $s = 1$ are computed (a). Next, the coefficients, when the wavelet with a translation $\tau = 1$, without changing the dilation are obtained (b). Likewise, for the same “ s ” of the wavelet, the coefficients, $C_1 \dots C_n$ are obtained. Next, the steps described in (a) through (c) are repeated for different dilations of the wavelet. Then, a complete scalogram depicting the time-scale representation of the signal under study is obtained.

Discrete Wavelet Transformation

Although the transformation obtained by shifting the wavelet continuously along the signal is called CWT, the transformation obtained by shifting the wavelet in discrete steps, with step-length equal to 2^j ($j = 1, 2, \dots, n$), is called DWT. It is important to note here that DWT will also be translational invariant (see subsection on Translational Invariance), only if the translations are in steps of 2^j ($j = 1, 2, \dots, n$), in which case, it is called dyadic wavelet transformation. In DWT, a discrete set of wavelet scales and translations are used. In fact, the DWT operation is similar to digital filtering. DWT is computed using Mallat’s algorithm (Mallat 1989), which essentially employs successive low-pass and high-pass filtering. First, the signal ($X[n]$) to be transformed is decomposed into high-frequency ($H[n]$) and low-frequency components

$(L[n])$ (also known as detailed and approximate coefficients, respectively). The detailed coefficients are known as level 1 coefficients. Then the approximate coefficients (low-frequency components) are again decomposed into the next level of detailed and approximate coefficients, and so on. Figure 1.3 depicts the DWT tree; a pictorial representation of the DWT process. At this point, it is important to understand exactly what happens at each level of decomposition when the signal under investigation is divided into low-frequency and high-frequency components. First, this enhances the resolution in frequency, thereby reducing its uncertainty by half. Accordingly, each of the filtered bands will have a band occupancy of $\omega/2$, which can be sampled at ω (cf. Nyquist theorem and its band pass extension). This decimation by 2 halves the time resolution as the entire signal after the first level of decomposition is represented by only half the number of samples. Thus, in the second level of decomposition, while the half-band low-pass filtering removes half of the frequencies, the decimation by 2 doubles the scale. With this approach, while the time resolution in the signal gets improved at high

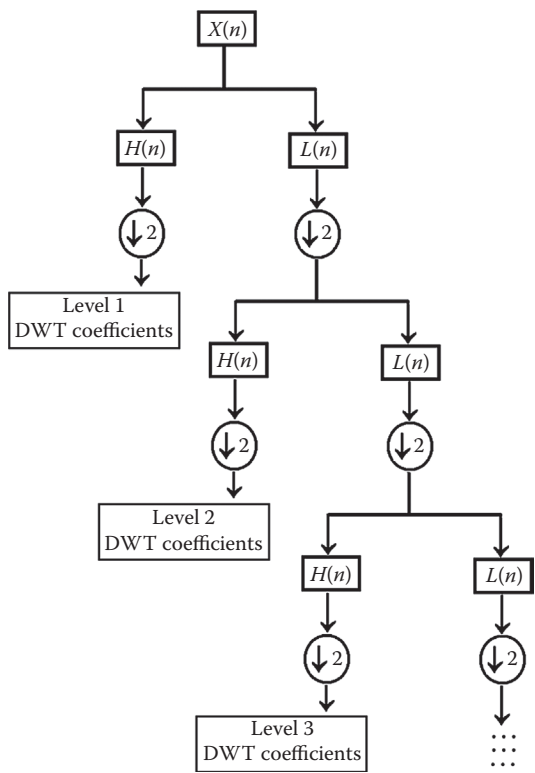


FIGURE 1.3
The DWT tree. The number “2” in the circles represents the decimation factor. See text for a detailed description of this tree.

frequencies, the frequency resolution is improved at low frequencies. Such a process is repeated until the desired levels of resolution in frequency are reached. However, the important point to note here is that at every wavelet operation, only half of the spectrum is covered. This means an infinite number of operations are needed to complete DWT. The solution to this problem does not lie in computing the DWT all the way down to zero with wavelet spectra, but to calculate up to a reasonable level, when it is small enough. The scaling function of the wavelet is used to check this. The scaling function is just a signal with a low-pass spectrum. The entire DWT operation can be summarized as follows: In DWT, if we analyze a signal using the combination of scaling function and wavelet function, the scaling function contributes to the calculation of the low-pass spectrum, whereas the rest is done by the wavelet function. The DWT of the original signal is then obtained by concatenating all the coefficients, $H[n]$ and $L[n]$, starting from the last level of decomposition. More details regarding DWT are given in Chapters 2, 3, and 4.

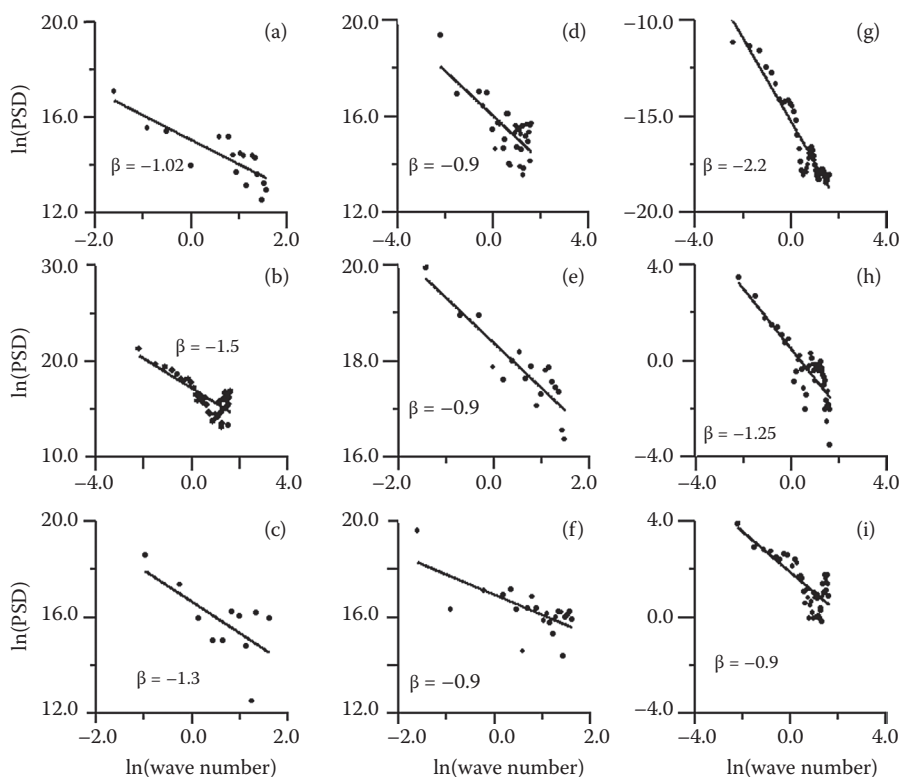
Fractals in Time Series Analysis

Often, geophysical data are in the form of time–space series (Dimri 1992). It has a combination of stochastic, trend, and periodic components (Malamud and Turcotte 1999). To know the stochastic component, one must compute the statistical distribution of values and persistence. The most commonly used method to determine persistence is spectral analysis, in which the power spectrum of a time series is plotted against the frequency (or wave number in case of a space series) and the value of a slope, known as the scaling exponent (e.g., β), giving an estimation of persistence (Dimri et al. 2012), which could be weak or strong. There exists a relation between β , the Euclidean dimension (E), and fractal dimension (D) of data as

$$D = E - 1 - \beta \quad (1.8)$$

Persistence of the time series is a measure of correlation between adjacent values of the time series. The uncorrelated or random data with zero persistence is the white noise or $\beta = 0$. The time series is antipersistent if adjacent values are anticorrelated, that is, $\beta < 0$. The time series is persistent if adjacent values are positively correlated with each other. For $\beta > 0$, the series is positively correlated.

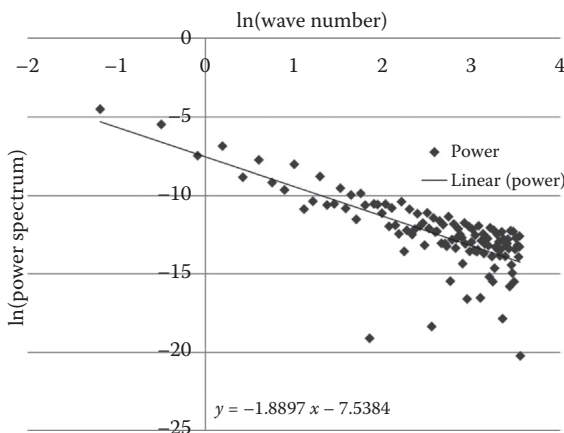
The density, susceptibility distribution, and reflectivity sequence of many boreholes around the globe have been investigated. The power spectral density of density, susceptibility distributions, and reflectivity sequences of some of the boreholes are shown to follow a fractal distribution (Dimri 2000a). The power spectrum of susceptibility distributions from 4 km depth of the German Continental Deep Drilling Project (KTB) was computed by

**FIGURE 1.4**

Log-log plot of wave number versus psd (in db) (a) for ODP Borehole complex impedance data; (b) of resistivity data for depth range one, KTB (VB); (c) of resistivity data for depth range two, KTB (VB); (d) of resistivity data for depth range three KTB (VB); (e) of resistivity data for depth range 1.6 to 2 km, of KTB (VB) a test case; (f) of resistivity data for depth range 2.0 to 2.5 km, of KTB (VB) a test case; (g) of conductivity data of KTB (HB); (h) of chargeability data for depth range one of KTB (VB); and (i) of chargeability data for depth range three of KTB (VB). (After Vedanti, N., and Dimri, V.P., *Indian J. Mar. Sci.* 32(4), 273–278, 2003.)

Maus and Dimri (1995a). It can also be observed in Figures 1.4 and 1.5 that there exists a fractal behavior in the electrical and thermal properties of the continental crust (Vedanti and Dimri 2003; Vedanti et al. 2011). From these investigations, it can be clearly seen that the power spectrum is not flat but rather it follows some scaling laws.

Now, once it is known that the physical properties follow a fractal behavior, it entails the reformulation of the mathematical formulae in frequency domain for existing techniques in geophysics that have been derived based on the Gaussian/random distribution of the physical properties. Such an approach was first developed by Spector and Grant (1970) and Naidu (1970), and was used to estimate the thickness of sedimentary basin from gravity and magnetic data. This method is known as the spectral analysis

**FIGURE 1.5**

Log-log plot of power spectrum of heat production data versus wave number for the Hidaka metamorphic belt (Hokkaido, Japan). Best linear fit gives a scaling exponent of 1.9, indicating the presence of fractal behavior in the data set. Nature of linear fit indicates strong correlation between the adjacent points and substantiate power-law regime in the data. (After Vedanti et al. 2001).

method and was also used by Negi et al. (1986) to estimate the thickness of the basalt in the Deccan Volcanic Province. However, for the fractal distribution of physical properties, a new method called scaling spectral method was developed and applied for various case studies of potential field data (Pilkington and Todoeschuck 1993; Maus and Dimri 1994, 1996; Fedi et al. 1997; Dimri 2000c; Bansal and Dimri 1999, 2005; and Bansal et al. 2006).

Different Methods to Estimate Fractal Dimensions

The essence of fractal analysis is the estimation of fractal dimension. With the use of fractal dimension, one can measure the length of a curve or area of rugged boundaries, or the volume of uneven surfaces, more precisely. Before the use of fractal dimensions, one was forced to use the Euclidean concept in which geometrical shapes consisting of straight-line polygons have the integer dimension values 1, 2, and 3 for a line, area, and volume, respectively. Mathematically, the length, L of a curve is a function of the size of the measuring instrument, for example, a scale of length ϵ . The relationship between L and ϵ is given by

$$L = A (1/\epsilon)^d \quad (1.9)$$

where A is a constant and d is the “measured fractal dimension.” Evaluating the logarithm of Equation 1.9 on both sides, the parameter, “ d ” can be obtained by

$$\log L = \log A + d \log(1/\epsilon) \quad (1.10)$$

Therefore, if we plot $\log L$ versus $\log (1/\varepsilon)$, we get a straight line whose slope gives the required value for d . One can show that the measured dimension, d , is related to the conventionally defined fractal dimension D by the relation:

$$D = 1 + d \quad (1.11)$$

If we calculate d for the coastline of Britain using the above procedure, we get a value of 0.28. Thus, we obtain the fractal dimension as 1.28. Therefore, the fractal dimension of the coastline of Britain is a fractional number (i.e., not an integer) between 1 and 2. The important point is that the fractal dimension is greater than the topological dimension (the topological dimension of a one-dimensional object is 1, in this case, the coastline). On the other hand, if we measure D for objects such as circles, we find that fractal dimension equals the topological dimension (both are equal to 1). Natural objects such as coastlines, exhibit fractal characteristics only over a limited range of scales, that is, the straight line behavior in the log-log plot used to calculate d holds only for a certain range of ε , that is, measuring length (Rangarajan 2000).

For complex cases such as heterogeneity in seismicity and fluid turbulence (Frisch and Parisi 1985), a single fractal dimension cannot explain the process. Such cases are best modeled using multifractals. Moharir (2000) demonstrated that the sum of two fractal processes is not a fractal. In such a case, a generalized fractal dimension, D_q , as a function of q , which varies from $-\infty$ to $+\infty$ was introduced. Hentshel and Procaccia (1983) presented D_q as

$$D_q = (1/q - 1) \lim_{r \rightarrow 0} \left[\log \left(\sum \{P_i(r)\}^q \right) / \log r \right] \quad (1.12)$$

where $P_i(r)$ is the probability that the events fall into a box with a size r . In the case of $q = 0, 1$, and 2 : D_0 coincides with the capacity dimension (D_0); D_1 with the information dimension (D_i) and D_2 , to the correlation dimension (D_{corr}). In general, the relation $D_0 > D_1 > D_2 > \dots D_\infty$ occurs within D_q . The multifractal analysis of complex phenomena involves estimation of the generalized dimension D_q first and the $D_q - q$ relation curve. There are various methods to determine D_q as given below.

Box Counting Method

It is a very popular method to estimate fractal dimension. In this method, the data are initially superimposed on a square grid of size r_0 . The unit r_0^2 is sequentially divided into small squares of size $r_i = r_0/2, r_0/4, r_0/8$, etc. The number of squares $N(r_i)$ intersected by at least one feature is counted each time. If the system is a self-similar structure, the relation between $N(r_i)$ and fractal dimension D is given by

$$N(r_i) \propto \left(\frac{r_0}{r_i} \right)^D \quad (1.13)$$

where D is interpreted as the fractal dimension of the lineament/drainage system. The fractal dimension D can be determined from the slope of the plot of $\log N(r_i)$ versus $\log (r_0/r_i)$.

Fixed-Size Algorithm

This is a generalization of the correlation method. In this method, the estimation of scaling of mass (e.g., number of earthquakes) within the circle of radius r increases with r . The dimension D_q is estimated from the scaling of mass with size for fixed-sized circles:

$$\log \langle M(<r)^{q=1} \rangle \approx (q-1) D_q \log r \quad (1.14)$$

Here, $M(<r)$ is the mass within the fixed radius, $\langle \rangle$ means the average of the mass for fixed size of the circle. The mass can be the number of data sets under consideration. The method is reported to be effective for determining the spectrum for $q > 0$ (Grassberger et al. 1988), but it is unstable for negative q when the data are limited.

Fixed-Mass Algorithm

This is also a generalization of the correlation method. The algorithm is the same as that used in the fixed radius method. Here, the smallest radius within which a fixed mass m can be included increases as the mass increases. The relationship to be used is

$$\log \langle R(<m)^{-(q-1)D} \rangle \approx -(q-1) \log m \quad (1.15)$$

The notations are similar to Equation 1.14. This method is reported to be superior to the previous one for calculating the spectrum for negative q (Grassberger et al. 1988). The applications for earthquake data are shown by several other workers (for example, see Hirata 1989; Hirabayashi et al. 1992).

Correlation Integral Method

In this method, the local density function $n_i(r)$ is defined by the following equation:

$$n_i(r) = \left(\frac{1}{N-1} \right) \left\{ \lim \sum (r - |x_i - x_j|) \right\} \quad (1.16)$$

where N is the number of data points recorded with a temporal resolution period, $(r - |x_i - x_j|)$ is the Heaviside function that counts how many pairs of points (x_i, x_j) fall within the scaling radius, r .

$$(r - |x_i - x_j|) = 0, \text{ if } (r - |x_i - x_j|) < 0 \quad (1.17)$$

$$= 1 \text{ if } (r - |x_i - x_j|) > 0 \quad (1.18)$$

To study the multifractal behavior of time series of earthquakes, r is replaced by scaling time t (Wang and Lee 1996) and $(|x_i - x_j|)$ by the inter-occurrence time $(|t_i - t_j|)$. Therefore, a generalized correlation integral $C_q(t)$ for the interoccurrence time t is defined by the equation

$$C_q(t) = \left\{ \left[\sum (n_i(t))^{q-1} \right] / N \right\}^{1/q-1} \quad (1.19)$$

Using this correlation integral, the temporal generalized fractal dimension D_q is then defined by the scaling relation as

$$Cq(t) \sim tDq \quad (1.20)$$

The generalized fractal dimension Dq is determined by plotting $Cq(t)$ versus t on a log-log graph. Applications of multifractals to earthquake data have been shown by Sunmonu et al. (2001) and Teotia (2000). There is a separate chapter on multifractals by Teotia and Kumar (2013).

Geophysical Significance of Wavelets and Fractals

Wavelets in Geophysics

Wavelet transformation has evolved due to the insufficient resolution that FT and WFT offered in determining the time–frequency localization in nonstationary signals.* Because the geophysical signals are, by nature, nonstationary, wavelets play a key role in understanding and interpreting the geophysical data more meaningfully. Jean Morlet, a French geophysicist, first noticed the importance of wavelets in the improved understanding of seismic signals when he provided a better interpretation of the subsurface geology by analyzing the backscattered seismic signals using wavelets

* These are the signals whose statistical properties, such as mean, standard deviation, and others, are time variant.

(Morlet 1981). Later, a thorough mathematical formalism for CWT was made (Goupillaud et al. 1984; Grossmann and Morlet 1984; Grossmann et al. 1989). Recognizing the important limited spatial support that wavelets offer, Farge (1992) employed wavelets to analyze turbulent flows. Wavelet studies of flow patterns later found many applications in solving problems of atmospheric turbulence (Katul et al. 1994; Narasimha 2007, to cite a few), climate studies (Lau and Weng 1995; Torrence and Compo 1998), and space–time rainfall studies (Narasimha and Kailas 2001; Labat et al. 2005; Bhattacharya and Narasimha 2007; Azad et al. 2008). Suitable applications have also been found in geomagnetism. Alexandrescu et al. (1995) and Adhikari et al. (2009) studied the sudden jumps (called geomagnetic jerks) observed in decadal variations of geomagnetic fields using wavelet analysis, as wavelet analysis is best suited for identifying the discontinuities or abrupt changes in the signals. Geomagnetic jerks, which usually have a time span of 1 year, arise due to the combination of a steady flow and a simple time-varying, axisymmetric, equatorially symmetric, toroidal zonal flow of the core fluid (Bloxham et al. 2002). Studies of geomagnetic jerks are important to understand lower mantle conductivity (Alexandrescu et al. 1999). A detailed study on the phase characteristics of geomagnetic jerks using complex wavelets is provided in Chapter 7 of this book (Chandrasekhar et al. 2013). More recently, Kunagu et al. (2013), while studying the characterization of external source fields, applied CWT to 10 years of CHAMP satellite magnetic data and identified some unmodeled signals in geomagnetic field models. Kunagu and Chandrasekhar (2013) later implemented the results of Kunagu et al. (2013) for geomagnetic induction studies. Moreau et al. (1997, 1999), Sailhac et al. (2000, 2009), and Sailhac and Gibert (2003) employed CWT to interpret the geophysical potential field data and to identify the sources of potential fields. Wavelet analysis also found important applications in the analysis and comprehensive understanding of geophysical well-log data, and provided space localizations of different subsurface formations. Wavelet analysis helped to estimate the preferential flow paths and the existence of flow barriers within the reservoir rocks (Jansen and Kelkar 1997), to determine high-frequency sedimentary cycles of oil source rocks (Prokoph and Agterberg 2000), to identify reservoir anomalies from pressure transient data (Panda et al. 2000; Soliman et al. 2001), for detection of cyclic patterns in well-log data (Rivera et al. 2004) and to the gamma ray log data to identify depths to the top of the formation zones (Choudhury et al. 2007). By making a histogram analysis of wavelet coefficients, Chandrasekhar and Rao (2012) optimized the suitable wavelet for studying the geophysical well-log data.

In this book, various other applications and developments in wavelet-based and fractal-based data analysis techniques in various processes of Earth system sciences, namely, paleoclimate studies, a study of non-stationary behaviour of rainfall anomalies using tree-ring data, phase field modeling studies and seismology and earthquake prediction studies using multifractal analyses, etc., are detailed in respective chapters.

Fractals in Geophysics

Various applications of fractals, particularly in the field of earth sciences, are given in the books by Mandelbrot (1983), Turcotte (1992), Feder (1988), Dimri (2000a, 2005a), Dimri et al. (2012), and Srivastava (2013; this volume). A collection of articles on geophysical applications is edited by Scholz and Mandelbrot (1989). Also, the application of fractals in a potential field survey design is given in Dimri (1998) and Srivastava et al. (2007). An application-oriented text on fractal application in reservoir engineering is given by Hardy and Beier (1994) and Dimri et al. (2012). Fractal theory has been extensively used in exploration geophysics, particularly in the interpretation of gravity and magnetic data (Pilkington and Todoeschuck 1993; Maus and Dimri 1994, 1995a,b, 1996; Bansal and Dimri 1999, 2005; Bansal et al. 2006; Fedi et al. 1997; Dimri and Srivastava 2005; Srivastava et al. 2009), and in seismic data (Tommy et al. 2001; Srivastava and Sen 2009, 2010). The application of monofractals and multifractals has been demonstrated in earthquake and tsunami studies [Teotia 2000; Teotia and Kumar 2013 (this volume); Sunmonu and Dimri 1999, 2000; Sunmonu et al. 2001; Mandal et al. 2005; Dimri 2005b; Dimri and Srivastava 2007]. Fractal theory can be applied in soil properties (Dimri 2000b; Ahmadi et al. 2011), biodiversity (Dimri and Ravi Prakash 2001), and chaotic studies (Dimri et al. 2011). Attempts have also been made to understand crustal heat production in the thermal regime of the continents (Dimri and Vedanti 2005; Vedanti et al. 2011). Those interested in the mathematical development of fractal theory are encouraged to follow Falconer (1990).

Geophysical data are often nonstationary or time-varying. There are various methods to convert the time-varying series into piecewise (time invariant) stationary series (Wang 1969; Dimri 1986; Bansal and Dimri 1999), as well as other methods such as Kalman filtering and adaptive approaches (Dimri 1992). Dimri and Srivastava (1987, 1990) used such an approach for seismic deconvolution problems. After the discovery of wavelets, a combined approach of wavelets and fractals for realistic geology studies was found to be very useful in a series of articles by Dimri et al. (2005), Chamoli and Dimri (2007), and Chamoli et al. (2006, 2007, 2010, 2011).

Relation between Wavelets and Fractals

By now, we understand the concepts of wavelets and fractals. As seen in the previous section, there are several methods to determine the fractal dimension (see Wang et al. 2003 for a review). In addition to the above methods, spectral methods (i.e., Fourier and wavelet methods) are also used to estimate the fractal dimension. McCoy and Walden (1996) and Wang et al. (2003) show that among the spectral methods, the wavelet method gives a better

estimate of fractal dimension than the Fourier method. In this section, we further discuss the relation between wavelets and fractals and the advantage of wavelet-based fractal analysis of nonstationary geophysical signals.

Let us consider the example of a Haar wavelet. Figure 1.6 explains that the scaling function, $\varphi(t)$, of the Haar wavelet can be expressed as a sum of its dilated and translated versions, for example, $\varphi(2t)$ (Figure 1.6a) and $\varphi(2t - 1)$ (Figure 1.6b). Mathematically, it is expressed as

$$\varphi(t) = \varphi(2t) + \varphi(2t - 1) \quad (1.21)$$

which gives the scaling function for a Haar wavelet (Figure 1.6c).

A close observation of these scaling functions clearly explains the self-similar nature of the Haar wavelet at different scales, which can be effectively employed for studying the self-similar nature of fractal images using the Haar wavelet. Wavelet transformation proves to be a very efficient tool for analyzing fractal objects. Chapter 4 of this book (and references therein) offers more details of such a description of multiresolution wavelet analysis of fractal images. Using the real Gaussian wavelet, Arneodo et al. (1988) computed the wavelet transform of the standard triadic Cantor set and showed that the wavelet transform allows us to capture the full complexity of the self-similar properties of multifractals. They also claim that the application of the wavelet transform to a variety of physical situations such as percolation, growth phenomena, and fully developed turbulence to name a few, looks very promising. Of late, the use of wavelets in analyzing fractal imaging has gained unprecedented importance. Davis (1998) provides a detailed explanation on the advantages of wavelet based analysis of fractal image compression. By comparing the conventional fractal coding algorithm based on an iterated function system (IFS) employed on the famous “Lenna” and “Baboon” images, and by developing a multiresolution fractal coding algorithm to analyze the above images using the Daubechies wavelet of order 2 (db2), Cesbron and Malassenet (1997) demonstrate the advantage of the latter technique in image compression methods. Recently, Lopez and Aldana (2007) adopted wavelet-based fractal analysis and waveform classifiers to determine fractal parameters from geophysical well-log data.

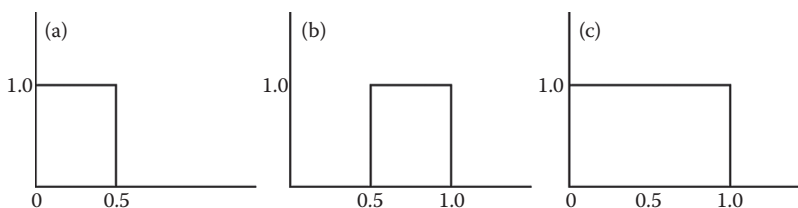


FIGURE 1.6

The scaling functions of Haar wavelet functions: (a) $\varphi(2t)$, (b) $\varphi(2t - 1)$, and (c) their sum, $\varphi(t)$, representing the self-similar nature of the wavelet at different translations and dilations.

Methodology for Wavelet-Based Fractal Analysis

For wavelet-based fractal analysis, first the variance of the wavelet coefficients is obtained at each level of decomposition. Then the log-variance is plotted against the scale. The linear portions in such plots correspond to a power-law process over a particular frequency region, indicating the fractal nature of the phenomena in that frequency range. The slope of the line can be related with the exponent of the power-law process (Percival and Guttorp 1994). Akay (1995) has employed the above methodology and determined the dimension of the echocardiatic signals. Using the log-variance of the wavelet coefficients obtained using db2 wavelet (which has self-similar characteristics), Huang and Morimoto (2008) determined the fractal dimension (also known as Hurst exponent) for the El Niño/La Niña episodes. In wavelet-based fractal analysis, because the relation between fractal dimension and scale is established, the fractal dimension obtained through this method, will in fact characterize the signal (a well-log signal or a seismic signal) that could be associated with different sedimentary formations. Enescu et al. (2004) studied the multifractal and correlation properties of real and simulated time series of earthquakes, using a wavelet-based multifractal analysis and quantitatively described the complex temporal patterns of seismicity, their multifractal and clustering properties in particular. Further research in various fields of science and engineering are actively progressing on the application of wavelet-based fractal analysis of nonlinear signals with wavelet coefficients obtained by two-dimensional wavelet transformation.

Conclusions

In this chapter, we have explained the fundamental concepts of wavelets and fractals and their advantages over conventional signal processing techniques in various problems from different fields of science and engineering research. We have also explained (by describing simple examples) and showed the need for these novel wavelet and fractal signal analysis tools to understand these signals in a better and wider perspective. Highlighting the role of wavelets, fractals, and multifractals—particularly in geosciences research—we have provided a number of examples in which these novel signal analysis tools have proven to be very effective. While discussing the relation between wavelets and fractals, we have highlighted the role of wavelets in fractal analysis of geophysical data as well as in image compression techniques. The reader is expected to read all the chapters of this volume and some more books and other literature to have a broader understanding of the subject for its further use in the analysis of a variety of nonstationary signals.

Acknowledgments

The authors thank two anonymous referees for their meticulous reviews, which have improved the quality of this chapter.

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2 Chapter 2: Construction of Wavelets: Principles and Practices

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(a) (b) (c) (d)

FIGURE 2.23

Fingerprint images reconstructed from level-5 DWT decomposition at a compression ratio of

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3 Chapter 3: Genesis of Wavelet Transform Types and Applications

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FIGURE 8.8

When the raw data is filtered for low frequencies using filter F3, periods of less than 5 years

appear highly accentuated. Effect of order (4 here) can be seen by comparing Figure 8.3 with

order 3. Axis descriptions are the same as in Figure 8.3.

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 3, 435-449. 40 30 30 20 10 20 10 -10 0 -20 -30 1950 1955
 1960 1965 1970 1975 1980 1985 1990 1995 1950 1955 1960 1965
 1970 1975 1980 1985 1990 1955 1960 1965 1970 1975 1980 1985
 1990 1995 Year 1955 1960 1965 1970 1975 1980 1985 1990 Year
 (a) (b) dY/dt (nT/yr) dY/dt (nT/yr) S c
 a l e 30 20 10 S c a l e Phase (degrees) Phase (degrees)
 Station: Lerwick (LER) Station: College (CMD) -100 1000
 -100 1000 1977 1969 1990 6 4 2 6 8 4 2 5 4 3 2 0 0 2 4 log
 2 (scale) 0 2 4 log 2 (scale) 0 2 4 log 2 (scale) l o g 2 |
 C W T τ , s | l o g 2 | C W T τ , s | $\alpha = 2.2$ 1970 6 4 2 0
 0 2 4 log 2 (scale) l o g 2 | C W T τ , s | $\alpha = 2.06$ $\alpha =$
 1.83 $\alpha = 1.45$

FIGURE 7.7

Phase scalogram plots together with secular variation plots
 and the ridge function plots related

to the 1969-1970 event corresponding to (a) LER and (b) CMD
 observatories of polar region. (a) (b) (c) 30 20 10 S c a l
 e 1945 1950 1955 1960 1965 1970 1975 1985 1990 1995 1980
 1945 1950 1955 1960 1965 1970 1975 1985 1990 1995 1980 Phase
 (degrees) -100 1000 Station: Eskdalemuir (ESK) 50 40 30 20
 10 dY/dt (nT/yr) Year Year 1969 1978 1989 6 4 2 0
 0 2 4 log 2 (scale) l o g 2 | C W T τ , s | 6 4 2 6 4 2 0 0
 2 4 log 2 (scale) 0 2 4 log 2 (scale) l o g 2 | C W T τ , s
 | $\alpha = 2.30$ $\alpha = 1.58$ $\alpha = 1.45$ 30 20 10 S c a l e 1955 1960
 1965 1970 1975 1985 1990 1980 1955 1960 1965 1970 1975 1985
 1990 1980 Phase (degrees) -100 1000 Station: Wingst (WNG) 50
 40 30 20 dY/dt (nT/yr) 1969 1978 1990 6 4 2 0 0 2
 4 log 2 (scale) l o g 2 | C W T τ , s | 6 4 2 3 4 2 0 0 2 4
 log 2 (scale) 0 2 4 log 2 (dilation) l o g 2 | C W T τ , s
 | $\alpha = 1.94$ $\alpha = 1.43$ $\alpha = 2.09$ Year 30 20 10 S c a l e 1955
 1960 1965 1970 1975 1985 1990 1980 1955 1960 1965 1970 1975
 1985 1990 1980 Phase (degrees) -100 -10 -5 0 1000 Station:
 Kakioka (KAK) dY/dt (nT/yr) 1969 6 4 2 0 2 4 log
 2 (scale) l o g 2 | C W T τ , s | $\alpha = 2.02$

FIGURE 7.8

Phase scalogram plots together with secular variation plots
 and the ridge function plots related

to the global jerk events corresponding to (a) ESK, (b)
 WNG, and (c) KAK observatories of the

high-latitude to mid-latitude regions. (a) (b) 30 20 10 S c
 a l e 1945 1940 1950 1955 1960 1965 1970 1975 1985 1990 1995

1980 1945 1940 1950 1955 1960 1965 1970 1975 1985 1990
 1995 1980 Phase (degrees) -100 1000 Station: Alibagh (ABG) d
 $Y / dt (nT / yr)$ Year Year 1951 1985 6 4 2 0 2 4 log 2
 (scale) log 2 | C W T τ , s | 6 4 2 0 2 4 log 2 (scale) log 2 | C W T τ , s | $\alpha = 2.08$ $\alpha = 1.32$ 30 20 10 S c a l e
 1965 1970 1975 1985 1990 1980 1965 1970 1975 1985 1990 1980
 Phase (degrees) -100 1000 Station: Bangui (BNG) 50 40 30 20
 10 d $Y / dt (nT / yr)$ 1971 1969 1982 1990 0 2 4 0 2 4
 0 2 4 0 2 4 log 2 (scale) log 2 (scale) log 2 (scale) log 2
 (scale) 6 4 2 8 0 6 4 2 8 0 6 4 2 8 0 6 4 2 8 0 1 0 g 2 | C
 W T τ , s | log 2 | C W T τ , s | $\alpha = 1.50$ $\alpha = 1.35$ $\alpha =$
 1.56 $\alpha = 2.15$ -30 -20 -10 0 10 20 (c) 80 75 70 65 60 55
 Year 30 20 10 S c a l e 1965 1970 1975 1985 1990 1980 1965
 1970 1975 1985 1990 1980 Phase (degrees) -100 1000 Station:
 Mbour S. (MBO) d $Y / dt (nT / yr)$ 1967 6 4 2 0 2 4 log
 2 (scale) log 2 | C W T τ , s | $\alpha = 1.50$ 1982 6 8 4 2 0 0
 2 4 log 2 (scale) log 2 | C W T τ , s | $\alpha = 2.15$

FIGURE 7.9

Phase scalogram plots together with secular variation plots
 and the ridge function plots related

to the global and local jerk events corresponding to (a)
 ABG, (b) BNG, and (c) MBO observato

ries of the mid-latitude to equatorial regions. (a) 30 20
 10 S c a l e 1955 1960 1965 1970 1975 1985 1990 1995 1980
 1955 1960 1965 1970 1975 1985 1990 1995 1980 Phase (degrees)
 -100 1000 Station: Hermanus (HER) d $Y / dt (nT / yr)$
 Year 1983 6 8 4 2 0 2 4 log 2 (scale) log 2 | C W T τ , s
 | $\alpha = 1.60$ -10 0 10 20 (c) 30 20 10 S c a l e 1965 1970
 1975 1985 1980 1965 1970 1975 1985 1980 Phase (degrees) -100
 1000 Station: Apia (API) d $Y / dt (nT / yr)$ Year 1976
 6 0 4 2 0 2 4 log 2 (scale) log 2 | C W T τ , s | $\alpha =$
 1.98 -10 0 10 20 (b) 30 20 10 S c a l e 1985 1990 1995 1980
 1985 1990 1995 1980 Phase (degrees) -100 1000 Station:
 Pamatai (PPT) d $Y / dt (nT / yr)$ Year 1980 6 0 4 2 0 2
 4 log 2 (scale) log 2 | C W T τ , s | $\alpha = 1.10$ 1985 0 4 2
 0 2 4 log 2 (scale) log 2 | C W T τ , s | $\alpha = 1.63$ 1989 0
 4 6 2 0 2 4 log 2 (scale) log 2 | C W T τ , s | $\alpha = 1.89$
 7 6 5 4 3 2 1 0

FIGURE 7.10

Phase scalogram plots together with secular variation plots
 and the ridge function plots

related to the local jerk events corresponding to (a) HER,
 (b) PPT, and (c) API observatories of

the southern hemisphere region.

(a)

(b) 0.08 0.07 0.06 0.05 0.04 0.03 0.02 0.01 Time (or space)

b

96

91

86

81

76

71

66

61

56

51

46

41

36

31

26

21

16

11

6

1

S c

a l e

s a

96

91

86

81

76

71

66

61

56

51

46

41

36

31

26

21

16

11

6

1

S c

a l e

s a 20 40 60 80 100 120 Time (or space) b 20 40 60 80 100
120 P e r c e n t a g e o f e n e r g y f o r e a c h w a v

e l e t c o e f f i c i e n t 0.04 0.035 0.03 0.025 0.02
 0.015 0.01 0.005 P e r c e n t a g e o f e n e r g y f o r
 e a c h w a v e l e t c o e f f i c i e n t Percentage of
 energy for each wavelet coefficient

FIGURE 9.3

(a) Wavelet spectrum of rainfall data without smoothening;
 (b) wavelet spectrum of rainfall

data after smoothening.